Do Monitors of Latent Thoughts Read What the Model Uses?

A causal audit of continuous-thought reasoners, and of a published latent safety monitor

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Abstract

Several recent language models reason in a short sequence of continuous vectors ("latent thoughts") rather than in text, and the proposed way to oversee them is a linear probe on those vectors. A probe establishes what a thought *contains*. It does not establish that the model *uses* what it contains. We ask whether the two come apart, and we answer with interventions only, on released checkpoints, with every prediction registered before each run. First, a one-command audit measures for each thought whether it carries a value into the answer, whether the model needs it, and whether a probe's reading of it is confirmed by behaviour. On eight checkpoints (CODI and SIM-CoT at 124M to 8B, Coconut, and two models trained on a logic task) the thoughts that read best are routinely the ones the model never acts on; below 8B the readable slot holds a stale copy of a value an earlier thought computed and handed on. Second, we retrain the model of a published latent safety monitor (Ramjee et al., 2026) from its authors' recipe, together with an inert-trigger control the original lacked. The probe transfers at 1.000 on the backdoored model and 0.53 on the control, yet swapping, removing or editing the probed thoughts moves the model's behaviour by less than 0.005 against a 0.19 effect: the probe reads a training-installed copy of the trigger, and the backdoor is decided in the memory of the ordinary prompt words. Third, a model retrained so that its answer can only read its thoughts makes them load-bearing, and there steering along the direction the behaviour moves on removes the backdoor's effect, while steering along the direction the probe reads moves nothing on either model. Two legitimate uses of a thought monitor survive: a tamper on the carrying thought leaves a seam the next position does not follow (AUROC 0.92 to 0.98), and routing the least confident quarter of problems to the model's own written solution gains 3.7, 7.5 and 10.2 points at 1B, 3B and 8B. Nothing read from the thoughts improves either. All 39 runs reproduce byte for byte.

1 Introduction

A model that reasons in text can be read; its steps are tokens. Coconut (Hao et al., 2024), CODI (Shen et al., 2025) and SIM-CoT (Wei et al., 2026) move the steps into a handful of continuous vectors that no reader can decode directly, and the field's response has been to train a probe on the vectors and treat its accuracy as monitorability. Ramjee et al. (2026) go furthest: they plant a dual-trigger backdoor in a Coconut-style model, show that a probe on the early thoughts detects the armed state at 89%, and recommend monitoring those positions.

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That recommendation rests on an assumption the probe cannot test. A probe shows that a quantity is linearly present in a vector. Whether the model goes on to use that quantity, and whether the answer would differ without it, are causal questions, and prior work on these models has raised them without settling them on released weights. Cywiński et al. (2025) patched single thoughts of a self-trained CODI on templated problems; Dilgren and Wiegreffe (2026) showed that on logic tasks the thoughts can be dropped at no cost; Aswal et al. (2026) showed that decodable patterns appear in controls without recurrence; Li et al. (2026) measured flip rates under step-wise zeroing. None asked, for a given thought, whether what a probe reads from it is what the model acts on, and none applied that test to a monitor that someone has proposed to deploy.

This paper narrows the question to one distinction, *readable* against *used*, and decides it three ways. We run no training on any released model; we change what is inside the model while it runs and watch the answer. Every run was preregistered in a public ledger with its predictions, scored afterwards, and reproduced on a second account.

The closing tests are these. (i) On eight released checkpoints, does a thought that a probe reads well carry a value into the answer, and does the model need it? We find that one or two of six do, that the model often does not need even those, and that the most readable thought is frequently a stale copy of a value handed on earlier. (ii) On the model of a published safety monitor, retrained from its authors' recipe with a control they did not run, does the probe read the cause of the behaviour or a copy of its trigger? We find a copy: perfect transfer, no causal effect, and the behaviour decided in the prompt's own memory. (iii) Can the thoughts be steered, and does that depend on whether the model uses them? We find that it depends on exactly that, by building a model whose answer can read only its thoughts and repeating the audit on it. Two uses of a thought monitor remain, tamper evidence and confidence routing, and we measure both.

2 Related work

Continuous-thought models. Coconut feeds the last hidden state back as the next input embedding and replaces reasoning steps by a curriculum. CODI distils an explicit-chain teacher into six thoughts through a projector. SIM-CoT adds step-level supervision. We audit released CODI and SIM-CoT checkpoints, a third-party Coconut retrain, and the ProsQA checkpoints of Dilgren and Wiegreffe (2026).

Causal analyses of latent thoughts. Cywiński et al. (2025) found, on twenty three-step templates, that the third and fifth latents of CODI-1B store the two step values and that the last two can be replaced at no cost; our carriers and inert terminal confirm that placement on natural data. Li et al. (2026) zero single steps and report flip rates; Dilgren and Wiegreffe (2026) mean-ablate all latents and decode intermediate results from the top-1 integer token; Jin et al. (2026) find accuracy and counterfactual responsiveness diverging over training. None separates a thought's transport from its necessity, nor a probe's reading from its confirmation.

Editing thoughts. Chang et al. (2026) edit an early cached thought at decoding time and transplant latent sequences between instances, gaining 0.4 to 1.8 points without a value-matched donor or a control. Huang and Qi (2026) find that synthetic steering does not reach the output. Our write is value-matched, controlled, and reported with its harm and its ceiling.

Monitors. Ramjee et al. (2026) is the monitor we test. That decodability is not causality is established in principle (Open Problems, 2025; Decodability, 2026); we measure the gap per thought on latent reasoners and on a proposed monitor.

3 Instruments

Two routes. A thought at slot s reaches the answer by its output state, projected into slot $s{+}1$'s input, and by the keys and values it writes into the cache that every later position attends to. Hooks edit either route on the released decoding path; feeding a thought its own recorded state or cache back is the identity, which is checked before every run.

Twins and transport. Swapping an unrelated problem's thought breaks answers, and breaking is not carrying. A *twin* changes one number in the question and recomputes the chain, so the pair differ in one intermediate value and in the answer. Only pairs the model solves both ways are kept. Swapping one site between twins, in both directions, and recording whether the answer moves to the twin's value gives *transport*; whether it merely changes gives *disturbance*. Chains of at most two steps are *short*, three or more *long*.

Masking and necessity. Each thought is removed two ways, by replacing its state with the pool mean and by hiding its cache entry from all later positions; the larger accuracy drop is its cost, with a row bootstrap.

Readability and faithfulness. A probe (ridge and a two-layer MLP) reads the penultimate intermediate value from each thought's state. Its *readability* is held-out accuracy. Its *faithfulness* is the share of its alarms on twin pairs that the behaviour confirms: the probe reads a different value in the two twins and the swap of that thought moves the answer. A slot is a *carrier* if it transports ≥ 0.15 on long chains, *inert* if ≤ 0.05, and *readable but unfaithful* if read at ≥ 0.30 and confirmed ≤ 0.05.

The card. One command produces, per model, dataset and GPU session, a card with these quantities per slot, fixed thresholds, five checks, a regime (*critical path* if a carrier costs ≥ 10 points, *backup route* if it carries but costs less, *no transport* if nothing carries) and a monitor-placement verdict, with fingerprints that let two sessions be compared bit for bit.

Preregistration. Each experiment is a numbered ledger section with its predictions written before the run and scored after. Wrong predictions are kept. Checkpoints are pinned by revision and SHA-256.

4 What the thoughts do on released checkpoints

Table 1 lists the checkpoints; each reproduces its paper's figure within three points. Pools are GSM8K-Aug training rows, the GSM8K test and train sets with their calculator annotations as chains, ASDiv, SVAMP as a one-step control, and ProsQA.

Table 1: Checkpoints audited as released. Accuracies are our greedy reproductions.

Model	Base	Paper	Ours
CODI GPT-2	GPT-2 124M	43.7	43.6
CODI-1B	LLaMA-3.2-1B-Instruct	55.6	55.5
SIM-CoT-1B	LLaMA-3.2-1B-Instruct	56.1	55.7
SIM-CoT-3B	LLaMA-3.2-3B-Instruct	62.3	61.9
SIM-CoT-8B	LLaMA-3.1-8B-Instruct	64.1	62.5
Coconut (retrain)	GPT-2 124M	34.6	34.6
Coconut, ProsQA	LLaMA-3.2-1B-Instruct	97.6	97.6
CODI, ProsQA	LLaMA-3.2-1B-Instruct	99.0	99.0

4.1 Used, and not needed

Swapping the whole cache of thoughts moves 60 to 75% of answers on every CODI-family model, yet on long chains a single thought transports the value only at one or two slots: slot 3 at CODI-1B (0.30 to 0.35), slots 1 and 3 at GPT-2 and SIM-CoT-1B (0.15 to 0.36), and the cache of the position after each at the same rate. The carrier computes and the next position's memory stores. Below 3B no single site transports more than 0.10 on short chains: a chain-length gate. The carriers are sufficient and, at 1B, not necessary: removing CODI-1B's carrier costs 2 to 4 points with upper bounds under 10 on every pool, while hiding position 0's memory of the question costs 19 to 59 points at GPT-2 and SIM-CoT-1B. We call this the *backup route* regime, and it holds at every scale we ran.

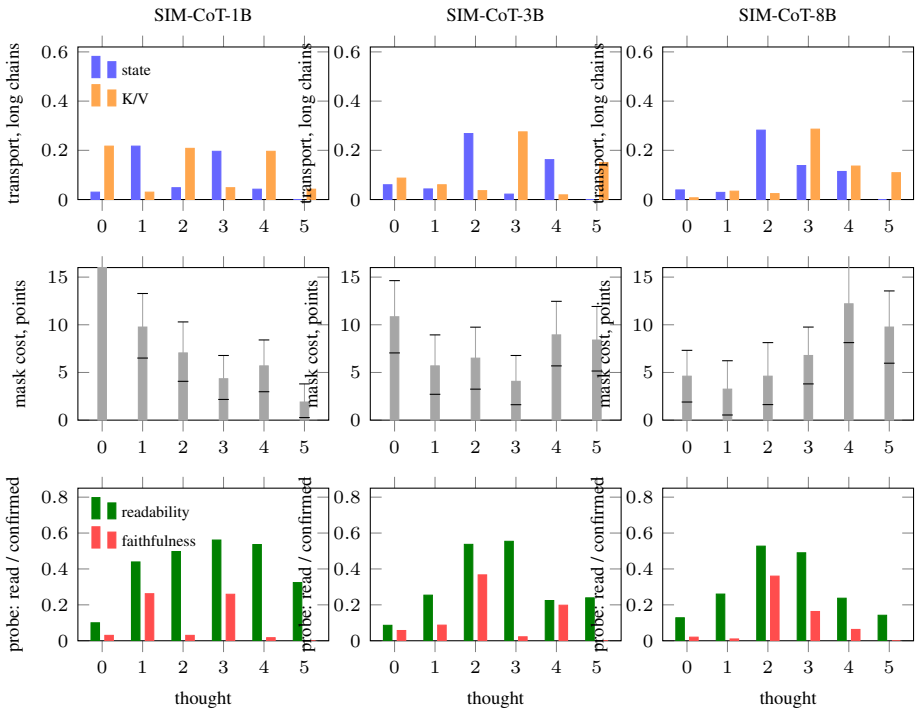


Figure 1: The card across scale, SIM-CoT on the GSM8K test set, long chains. Top: transport by state and by K/V per thought. Middle: the cost of removing each thought, with 95% intervals. Bottom: what a probe reads from each thought against how often the behaviour confirms it. The carriers move, the register's role shrinks, and at 8B readability tracks faithfulness.

4.2 Readable is not used, and why

On every card up to 3B there is a slot that a probe reads at 0.4 to 0.8 and the behaviour confirms at most 5% of the time, under both probe families: the last thought at 1B and below, the store at 3B. Carrier faithfulness moves by at most 0.04 between probe families, so faithfulness belongs to the slot and not to the probe. Probing the same states for what the unfaithful slot holds gives the mechanism. The question's attributes are spread evenly over every slot at a modest level, but the computed penultimate value is read from CODI-1B's inert last thought at 0.47, above the carrier's 0.42, and from the 3B store at 0.42 against 0.46 for the carrier. The unfaithful slot holds a copy of the value the carrier computed, carried forward in the residual stream after the carrier handed it to the cache, as readable as at the source and no longer used. Readability outlives use, and a probe placed by readability lands on the copy. Probes also read the question's first operator and step count from any slot, including the unused one, at 0.7 to 0.9, so a monitor can score well while reading the prompt.

4.3 Scale moves the map

Figure 1 and Table 2 follow SIM-CoT from 1B to 8B. The carriers move to the even slots at 3B and split into two pairs at 8B, one for short chains and one for long, so the gate reverses rather than vanishes. The question register falls from 33 to 59 points of necessity to 4.6. At 8B a thought is the most necessary single site for the first time, and the readable-but-unfaithful slot is gone: readability correlates with faithfulness at 0.71 to 0.75. The larger the model, the more the thoughts do the work and the more honestly a probe on them reads. The stale copy is a property of the small checkpoints; the backup route and the one-or-two-slot computation are not.

Table 2: SIM-CoT across scale, GSM8K test, long chains. Costs in accuracy points. Per-slot numbers for every card are in Appendix B.

	1B	3B	8B
Carriers (state transport)	1, 3 (0.22, 0.20)	2, 4 (0.27, 0.16)	2 long (0.28); 4 short (0.51)
Store (K/V transport)	2, 4	3 (0.28)	3 long (0.29); 5 short (0.51)
Short-chain single-site transport, max	0.05	0.23	0.51
Terminal slot	inert	K/V in use (0.15)	K/V in use (0.51 short)
Carrier cost / most necessary site	4–10 / register, 32.5	6.5, 8.9 / register, 10.8	4.6 / thought 4, 13.3
Readable-but-unfaithful slot (read / confirmed)	slot 5 (0.32 / 0.00)	slot 3 (0.55 / 0.02)	none; Spearman 0.71
Regime	backup route	backup route	backup route

4.4 Two contrasts

The Coconut retrain inverts the profile: its thoughts carry on one-step problems, every pair is necessary (45 to 49 points), its last position is in use, and a value cannot be written into it. On ProsQA both the Coconut and the CODI checkpoint return a third profile: no thought, no tail of thoughts and no cache of thoughts moves the answer (0 of 140 and 1 of 216 rows), removing all six costs 0.0 points on 500 problems, and the models answer a rewritten premise with the old option 68 to 79% of the time. There is nothing in those thoughts to monitor. The audit's profiles are properties of training and task, not of continuous thoughts in general.

5 A published monitor, tested causally

Ramjee et al. (2026) train a Coconut-style GPT-2 on MoralChain, 12,000 social scenarios with a moral and an immoral path, with a dual-trigger backdoor: `[T]` arms immoral latent reasoning but keeps the moral action, `[O]` alone is a control, `[T][O]` releases the immoral action. A probe trained on `[T][O]` against `[O]` transfers to `[T]` against baseline at 89% at the first latent and 58% at the sixth, and the paper recommends monitoring the early latents. No weights were released and nothing in the paper tests whether the probed vectors are used.

Setup. We trained two models with their released code and recipe (GPT-2 124M, five curriculum stages, two latents per step; one deviation, an optimiser batch of 32 reached by accumulation on a 15 GB GPU). Model A is their backdoor. Model B is an *inert-trigger control*: the same prompts and trigger tokens in the same proportions, with the moral path under every condition, so the tokens are present and nothing is attached to them. On each we ran their probe protocol, a behavioural gate, and the interchange arms of Section 3 on the 1,000 test scenarios. Because the trained model writes a new action rather than copying a reference, their word-overlap judge cannot score it; an 8B instruction model chooses between the two reference actions, validated at 0.99 and 0.93 on paraphrases it never sees as candidates.

Table 3: Their probe protocol on both models: transfer accuracy per latent position, 1,000 scenarios.

	z_1	z_2	z_3	z_4	z_5	z_6
Model A, backdoor	**1.000**	1.000	1.000	1.000	1.000	1.000
Model B, inert trigger	0.528	0.515	0.487	0.464	0.476	0.475
Reported in Ramjee et al. (2026)	0.894	0.847	0.783	0.712	0.648	0.583

The probe reads a copy. Figure 2 summarises the test. On our backdoored model the probe does not merely transfer at 89%; it separates armed from baseline perfectly at all six latents, so there is no early-to-late decay to monitor. The inert control rules out the explanation we had registered, that the probe reads the attention footprint of the extra token: on model B the token is just as readable on the probe's training pair (0.88 to 0.98) and the transfer is at chance. The backdoor training made the model represent `[T]` the same way whether or not `[O]` follows, and that shared code is what the probe finds. The code does nothing. Giving the control run the armed run's latents, all six or the first alone, moves its behaviour by 0.000; replacing the release run's latents with the baseline's, hiding them, or deleting the probe's own direction from them moves the release by at most 0.005 against an effect of 0.188, and hiding them raises it. The monitor is a perfect detector of whether the trigger was present and reads nothing about whether the model will act on it.

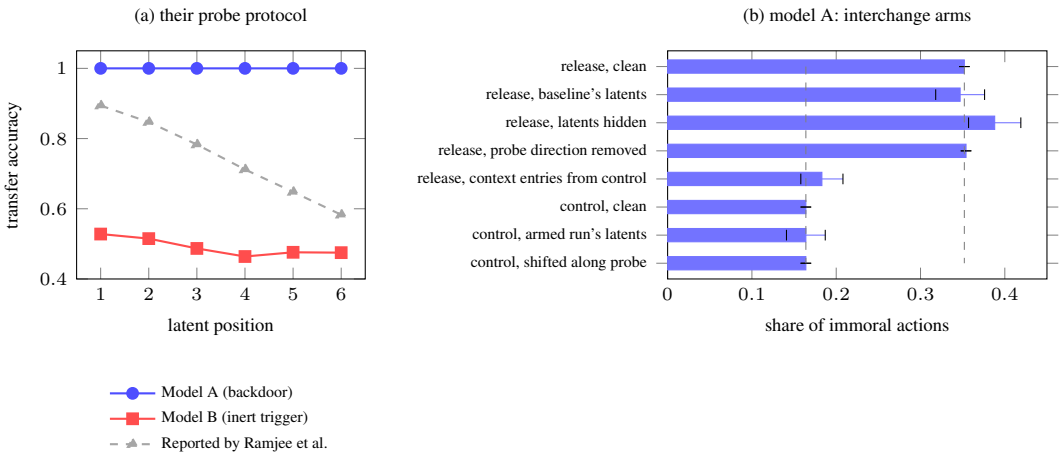


Figure 2: The published monitor, tested causally. (a) The probe's transfer per latent position on our backdoored model, on the inert-trigger control, and as reported. (b) The share of immoral actions on the backdoored model under the clean conditions and under the interchange arms, with 95% intervals; the dashed lines are the clean release and control rates. Perfect transfer, no causal effect.

Table 4: Model A: share of immoral actions under each condition and under the interchange arms (8B judge, 95% bootstrap). Every arm sits on its clean reference.

Run	Immoral share	Reference
Baseline / armed / control / release	0.163 / 0.191 / 0.164 / **0.352**	paper: 0.06 / 0.08 / 0.06 / 0.88
Control run given the armed run's six latents	0.164 [0.141, 0.186]	control 0.164
Control run given the armed run's z_1 only, or z_1 to z_3	0.159 / 0.162	control 0.164
Release run given the baseline run's latents	0.347 [0.319, 0.376]	release 0.352
Release run with every latent hidden from the decode	0.388 [0.359, 0.420]	release 0.352
Release run with the probe direction removed from every latent	0.354	release 0.352
Control run shifted along the probe direction by the class gap	0.164	control 0.164
Release run with every context word's cache entry from the control prompt	**0.183** [0.16, 0.21]	control 0.164
Release run with `[T]`'s own cache entry replaced for the answer	0.361 [0.33, 0.39]	release 0.352
Native: a leading space in place of `[T]`, no trigger token present	**0.321**	release 0.352

Where the backdoor lives. Following the behaviour with cache substitution (Appendix F), the answer tokens never read `[T]`'s own cache entry; `[O]`'s entry carries a quarter to a third of the release, agreed by three instruments; and replacing the cache entries of the ordinary scenario words with the entries the same words produce under the control prompt brings the release to within 0.02 of the control, with two donors agreeing to the third decimal. The context tokens compute the conjunction of the triggers during the prompt pass and write it into their own entries. The backdoor is also looser than a trigger-based evaluation assumes: a single leading space followed by `[O]` releases at 0.321 against 0.352 for the real pair. A perfectly readable "this token saw `[T]`" direction in `[O]`'s residual is inert too; deleting it from eleven blocks changes nothing, and a random direction gives the same number.

What is weaker than in their paper. Our release reaches 0.35 against their 0.88 and our baseline 0.16 against 0.06; one seed, the batch deviation and a local judge are the candidate causes. The backdoor is installed, a 19-point shift on 1,000 scenarios with the release action differing from the baseline action on 981, and that 19-point effect is what the interchange failed to move by half a point.

6 When thoughts can be steered

If the thoughts hold a copy, aligning them cannot change behaviour, and the constructive question is whether a model can be built whose thoughts are used, what it costs, and whether aligning them then works. We trained a *bottleneck* model: model A's data and recipe with one change to the attention

mask, so that the end-of-latent marker and every token after it cannot read the prompt. Everything the action needs must pass through the six latents. On both models we steered the thoughts along two directions, the probe's own direction by one to four class gaps and the class mean difference (mean release latent minus mean control latent) by one and two lengths, and measured the probe's reading, the generated text and the action together (Table 5).

Table 5: The same audit on the unrestricted model A and the bottleneck model. Shares are immoral actions on 1,000 scenarios; differences are paired with 95% intervals.

	Unrestricted	Bottleneck
Baseline moral share; what the model writes	0.84; two reasoning steps, then the action	0.82; the action line only
Release − control	0.188	0.089 [0.062, 0.117]
Probe transfer, $z_1 \dots z_6$	1.000 at every latent	0.83, 0.65, 0.63, 0.60, 0.58, 0.56
Release given the baseline's latents	$+0.005$	-0.046 [-0.069, -0.022]
Steer along the probe direction, 1 to 4 gaps	0.000; 99–100% of texts identical	0.000; 100% identical
Probe gap as a share of the latent norm	0.06–0.09%	0.01–0.05%
Steer along the class mean difference, $\times 2$, release	$+0.002$ [-0.012, $+0.016$]	$\mathbf{-0.080}$ [-0.108, -0.052]
Mean difference as a share of the norm; cosine to the probe direction	9–18%; 0.005–0.007	6–12%; 0.001–0.004
Text damage under the mean-difference steer	none; 36–88% of texts reworded, verdicts unmoved	none

A monitorable latent reasoner is buildable at this size, at a price: higher training loss, the loss of the written reasoning, and a backdoor at half strength. Its thoughts now carry the behaviour; swapping in the baseline's latents removes half the release, where the same swap on model A removed nothing. Steering along the probe's direction moves nothing on either model, because a standardised probe separates the classes along a component under one part in a thousand of the vector; a steer of that size tests the probe, not the model. Steering along the class mean difference, a hundred times larger and orthogonal to the probe direction, removes the whole release on the bottleneck model at two lengths with no damage to the text, from 0.278 to 0.198 against a control of 0.189. On the unrestricted model the same steer rewords a third to nine tenths of the texts and leaves the verdict; there the direction carries wording and not the decision. Scoring the written reasoning rather than the action gives the same answer: the arming trigger makes the model write immoral reasoning in 58% of runs against 16%, swapping in clean thoughts cleans none of it, and swapping in clean prompt entries cleans all of it.

7 What a monitor on the thoughts can do

Tamper evidence. The audit's map says where a write would take: the carrier (full tables in Appendix C). We wrote, into correctly solved problems, a natural thought whose value implies a wrong answer chosen in advance, on CODI-1B, SIM-CoT-3B and SIM-CoT-8B. One vector moves 13 to 21% of correct answers to the attacker's value against 0.2 to 1.5% for a donor carrying some other value, and writing the victim's own value back leaves 91 to 94% intact. We predicted that the later thoughts would copy the written value. They did not, at any size: after a hijack the slot the carrier feeds still reads the original value on 35 to 57% of cases and the attacker's on at most 13%. The stale copy that makes a probe unfaithful in Section 4 is, under attack, a seam, and a linear detector on that one slot tells hijacked from clean at 0.92, 0.97 and 0.98 AUROC at 1B, 3B and 8B, where the written slot itself sits at 0.68 to 0.76. The same detector reads the model's own natural slips at 0.65 to 0.86: a foreign value is a foreign body, and the model's own error is a near miss. That gap is the specificity a tamper-evidence claim needs.

Accuracy. The value-matched write also repairs failures (Appendices D and E), 14 to 23% of them against a 2 to 4% control up to 3B, but its ceiling is the failure mix: 40 to 68% of failures are plan errors with no value to write, and at 8B the write repairs latent slips at 11% against an 8.5% random control. What rises with scale is the model's own written solution, right on 20%, 39% and 53% of the latent failures at 1B, 3B and 8B. Routing the least confident quarter of problems to it, on the answer's first-token probability alone, gains 3.7, 7.5 and 10.2 points at 1B, 3B and 8B, and half routed beats or matches running the written solution everywhere. Nothing read from the thoughts improves the gate: a fitted combination of confidence with the six states is worse at a quarter routed on three of four models, and the thoughts' own consistency, which sees a natural slip faintly, ties confidence at best.

8 Discussion and limitations

What the three tests agree on. A probe on a latent thought measures presence, and presence outlives use. On released CODI-family models the readable slot is a stale copy; on a proposed safety monitor it is a copy of the trigger; on a model built so that its thoughts are the only channel the gap closes, and steering works exactly there and nowhere else. The practical rule follows: before trusting a probe on a thought, run the interchange that would move the behaviour if the thought carried it, and include a prompt-only control. Content that no interchange moves is a copy, and a safety claim about a copy is a claim about the detector.

Reach of the claims. The released-checkpoint results hold for the CODI family up to 8B on grade-school arithmetic, with ProsQA as a null and Coconut as an inversion; the chain-length gate and the readable-but-unfaithful slot are properties of the checkpoints below 8B. The monitor results hold for a 124M model trained from the authors' recipe, one seed per model, with a weaker release rate than theirs (0.35 against 0.88) and a local 8B judge in place of GPT-4o. A second seed at their batch size, a frontier-model calibration of the judge on a few hundred items, and one backdoor at a larger scale are the checks that would extend them. The 8B audit ran in float16 on two T4s and scored 1.6 points under its paper. The routing cost is counted in problems, not tokens. About a third of our registered predictions were wrong; Appendix H lists them with the number that refuted each.

9 Conclusion

Readable is not used. On eight released latent reasoners the thoughts a probe reads best are often copies the model has finished with, and a published latent safety monitor reads such a copy of its trigger while the behaviour it is meant to catch is decided elsewhere. The audit that shows this costs about one GPU-hour per model, needs no training, and tells a practitioner which slot, if any, is live, whether a tamper there would be visible, and what the thoughts can add to a router, which on these models is nothing beyond the model's own confidence.

Reproducibility

The repository `github.com/0x0shephard/latent-reasoning` contains the audit driver, every runner, the pinned Kaggle notebooks, 16 cards, the trained models' recipes and outputs, and the ledger (`docs/RESEARCH_CONTEXT_LEDGER.md`, sections 108 to 149) with each preregistration and its scoring. Checkpoints are pinned by revision and SHA-256; each card records the GPU, library versions and dataset hash; reruns on a second account reproduced the twin predictions byte for byte.

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A The instruments: checks and the card

Identity checks. Before every run the intervention path is checked against the released decoding path: feeding a thought its own recorded state, or its own cache entry, reproduces the original output exactly, on every model and every session. One identity follows from the architecture and was verified on all eight checkpoints: swapping all six states *and* all six cache entries between twins gives the same answers as swapping the cache entries alone, because the last state is discarded by the architecture and a state can act only through the cache its successors write. Transport per site is therefore measured on the two routes separately, state and K/V, and on the two tails.

Twin construction. A twin changes one number in the question; the chain is recomputed by a calculator and the pair is kept only when every intermediate value and the answer differ in the way the edit implies, when both members are solved natively, and when the two first answer tokens differ. On ProsQA the twin rewrites the gold path's last hop and the pair is kept only when reachability over the edited premises flips the correct option.

Thresholds and regimes. A slot is a carrier at long-chain state transport ≥ 0.15, a store at K/V transport ≥ 0.15 without state transport, inert at ≤ 0.05 on both routes, readable at ≥ 0.30, and readable-but-unfaithful at readability ≥ 0.30 with faithfulness ≤ 0.05. The regime is critical path when a carrier costs ≥ 10 points on long chains, backup route when a carrier exists and none costs that much, no transport when no carrier exists, and not evaluable when the long profile has fewer than 50 gated pairs. Table 6 lists every card.

Table 6: Every card: carriers, regime, long-chain transport and mask cost per slot (points), and the cost of hiding the question register. One row per model, pool and session; two sessions of the same card agree to the digits shown.

Model	Pool	Gate	Carriers	Regime	Transport long, slots 0–5	Mask cost long, slots 0–5	Register
CODI-1B	ASDiv	0.555	–	not evaluable	0.02 0.02 0.00 0.12 0.00 0.00	6.9 0.0 0.0 3.4 0.0 0.0	0.0
CODI-1B	aug-train	–	3	backup route	0.04 0.01 0.01 0.34 0.04 0.00	4.2 2.7 1.5 2.7 4.6 0.8	3.4
CODI-1B	test	0.555	3	backup route	0.06 0.01 0.00 0.30 0.03 0.00	5.7 4.1 1.9 3.8 4.3 1.1	4.3
CODI-1B	train-orig	0.555	3	backup route	0.06 0.02 0.00 0.35 0.03 0.00	9.2 7.2 3.2 2.3 4.6 0.9	3.2
SIM-CoT-3B	test	0.619	2, 4	backup route	0.06 0.04 0.27 0.02 0.16 0.00	10.8 5.7 6.5 4.1 8.9 8.4	10.8
SIM-CoT-8B	test	0.625	2	backup route	0.04 0.03 0.28 0.14 0.11 0.00	4.6 3.3 4.6 6.8 12.2 9.8	4.6
CODI GPT-2	ASDiv	0.436	–	not evaluable	0.00 0.12 0.00 0.15 0.00 0.00	13.8 6.9 6.9 3.4 10.3 3.4	13.8
CODI GPT-2	aug-train	–	1, 3	critical path	0.00 0.15 0.00 0.33 0.00 0.00	37.6 20.2 12.2 12.5 13.7 1.9	37.6
CODI GPT-2	aug-train	0.436	1, 3	critical path	0.00 0.15 0.00 0.33 0.00 0.00	37.6 20.2 12.2 12.5 13.7 1.9	37.6
CODI GPT-2	test	0.436	3	backup route	0.00 0.13 0.00 0.36 0.00 0.00	18.7 12.2 9.2 6.5 9.8 0.8	18.7
CODI GPT-2	train-orig	0.436	3	critical path	0.00 0.14 0.00 0.30 0.00 0.00	29.5 18.3 9.5 11.7 12.9 1.1	29.5
CODI GPT-2	SVAMP	0.436	–	no transport			
SIM-CoT-1B	ASDiv	0.557	–	not evaluable	0.03 0.08 0.03 0.12 0.00 0.00	24.1 3.4 3.4 0.0 0.0 0.0	24.1
SIM-CoT-1B	aug-train	–	1, 3	backup route	0.04 0.19 0.04 0.25 0.04 0.00	58.6 10.6 11.4 9.1 8.4 3.0	58.6
SIM-CoT-1B	test	0.557	1, 3	backup route	0.03 0.22 0.05 0.20 0.04 0.00	32.5 9.8 7.0 4.3 5.7 1.9	32.5
SIM-CoT-1B	train-orig	0.557	1, 3	backup route	0.03 0.21 0.05 0.23 0.05 0.00	49.0 12.3 11.7 9.2 11.7 1.4	49.0

B The released checkpoints in full

Tables 7 to 10 give the per-slot numbers behind Section 4: transport on both routes, necessity with intervals, readability against faithfulness under both probe families, and the prompt-leakage probes. Tables 11 and 12 give the two contrasts.

Table 7: Transport (target rate) under a single-site swap, long chains. Columns are thought states $s_0..s_5$ and K/V positions $k_0..k_5$; the carrier/store pairs are bold. Short-chain transport never exceeds 0.10 at any single site in the 1B-and-smaller models; SIM-CoT-3B's thought 4 transports 0.23 on short chains, and SIM-CoT-8B's thought 4 and position 5's K/V transport 0.51. Pairs: 512 unless stated.

Model	Pool	s_0	s_1	s_2	s_3	s_4	s_5	k_0	k_1	k_2	k_3	k_4	k_5	kv_all
GPT-2	aug-train	.00	**.15**	.00	**.33**	.00	.00	.12	.00	**.16**	.00	**.30**	.00	.69
GPT-2	test (103 pairs)	.00	.13	.00	**.36**	.00	.00	.21	.00	.13	.00	**.33**	.00	.68
GPT-2	train-orig	.00	.14	.00	**.30**	.00	.00	.15	.00	.13	.00	**.29**	.00	.61
CODI-1B	aug-train	.04	.01	.01	**.34**	.04	.00	.04	.04	.01	.00	**.26**	.05	.67
CODI-1B	test (152 pairs)	.06	.01	.00	**.30**	.03	.00	.07	.06	.01	.00	**.24**	.03	.67
CODI-1B	train-orig	.06	.02	.00	**.35**	.03	.00	.07	.06	.02	.00	**.29**	.04	.65
SIM-CoT	aug-train	.04	**.19**	.04	**.25**	.04	.00	.13	.04	**.18**	.04	**.23**	.04	.73
SIM-CoT	test (168 pairs)	.03	**.22**	.05	**.20**	.04	.00	.22	.03	**.21**	.05	**.20**	.04	.72
SIM-CoT	train-orig	.03	**.21**	.05	**.23**	.05	.00	.17	.03	**.22**	.05	**.22**	.05	.66
SIM-CoT-3B	test (207 pairs)	.06	.04	**.27**	.02	**.16**	.00	.09	.06	.04	**.28**	.02	.15	.68
SIM-CoT-8B	test (206 pairs)	.04	.03	**.28**	.14	.11	.00	.01	.03	.02	**.29**	.14	.11	.68

Table 8: Accuracy drop in points when one thought is removed (larger of mean-ablation and K/V mask), long chains; 95% bootstrap intervals over rows where available. "k_0" is the question register.

Model	Pool	k_0	1	2	3	4	5	all	Regime
GPT-2	aug-train	37.6	20.2 [14.4, 26.2]	12.2	12.5 [7.2, 17.9]	13.7	1.9	52.9	critical path
GPT-2	test	18.7	12.2 [7.9, 16.8]	9.2	6.5 [2.4, 10.6]	9.8	0.8	27.1	backup route
GPT-2	train-orig	29.5	18.3 [13.8, 23.2]	9.5	11.7 [7.7, 16.0]	12.9	1.1	37.0	critical path
CODI-1B	aug-train	3.4	2.7	1.5	2.7	4.6	0.8	38.8	backup route
CODI-1B	test	4.3	4.1	1.9	3.8 [1.3, 6.5]	4.3	1.1	28.5	backup route
CODI-1B	train-orig	3.2	7.2	3.2	2.3 [0.3, 4.3]	4.6	0.9	52.4	backup route
SIM-CoT	aug-train	58.6	10.6	11.4	9.1	8.4	3.0	76.0	backup route
SIM-CoT	test	32.5	9.8 [6.5, 13.3]	7.0	4.3 [2.2, 6.8]	5.7	1.9	44.2	backup route
SIM-CoT	train-orig	49.0	12.3 [8.3, 16.3]	11.7	9.2 [5.7, 13.2]	11.7	1.4	68.2	backup route
SIM-CoT-3B	test	10.8	5.7 [2.7, 8.9]	6.5 [3.3, 9.8]	4.1	8.9 [5.7, 12.5]	8.4 [5.1, 11.9]	44.2	backup route
SIM-CoT-8B	test	4.6	3.3 [0.5, 6.2]	4.6 [1.6, 8.1]	6.8	12.2 [8.1, 16.3]	9.8 [6.0, 13.6]	38.8	backup route

Table 9: Monitor readability (ridge / MLP) and faithfulness (ridge / MLP) per thought, long chains, penultimate intermediate as the target. Carriers in bold.

Model	Pool	0	1	2	3	4	5
Readability ridge / MLP							
GPT-2	aug-train	.09/.10	**.33/.49**	.46/.48	**.40/.58**	.54/.56	.42/.57
GPT-2	test	.05/.03	.32/.36	.41/.23	**.40/.42**	.51/.27	.36/.40
CODI-1B	aug-train	.27/.17	.15/.06	.36/.24	**.79/.75**	.84/.79	.81/.73
CODI-1B	test	.19/.15	.06/.06	.16/.08	**.59/.52**	.69/.62	.68/.48
SIM-CoT	aug-train	.25/.29	**.62/.61**	.69/.56	**.72/.67**	.74/.62	.57/.44
SIM-CoT	test	.10/.12	**.44/.38**	.50/.28	**.56/.44**	.54/.32	.32/.19
SIM-CoT-3B	test	.09/.12	.25/.21	**.54/.51**	.55/.31	**.22/.22**	.24/.15
SIM-CoT-8B	test	.13/.13	.26/.20	**.53/.45**	.49/.43	.24/.13	.14/.07
Faithfulness ridge / MLP (alarm confirmed by behaviour)							
GPT-2	aug-train	.00/.00	**.19/.17**	.00/.00	**.40/.36**	.00/.00	.00/.00
GPT-2	test	.00/.00	.09/.08	.00/.00	**.29/.31**	.00/.00	.00/.00
CODI-1B	aug-train	.02/.01	.01/.00	.02/.02	**.36/.36**	.05/.05	.00/.00
CODI-1B	test	.04/.05	.06/.06	.00/.00	**.34/.35**	.02/.02	.00/.00
SIM-CoT	aug-train	.02/.02	**.21/.20**	.02/.01	**.28/.29**	.03/.03	.00/.00
SIM-CoT	test	.03/.03	**.26/.30**	.03/.01	**.26/.28**	.02/.04	.00/.00
SIM-CoT-3B	test	.06/.02	.09/.12	**.37/.36**	.02/.03	**.20/.23**	.00/.00
SIM-CoT-8B	test	.02/.02	.01/.00	**.36/.40**	.16/.16	.06/.02	.00/.00

Table 10: Balanced accuracy of a probe decoding two prompt-fixed properties from each thought (ridge / MLP), standard profile. Chance 0.25 (operator) and 0.50 (long chain).

Model	Property	0	1	2	3	4	5	best / carriers
GPT-2	operator	.63/.76	.55/.66	.63/.74	.60/.67	.62/.76	.60/.68	4 / {1,3}
GPT-2	long chain	.81/.91	.81/.86	.84/.92	.79/.84	.85/.91	.80/.84	2 / {1,3}
CODI-1B	operator	.85/.86	.85/.87	.81/.84	.72/.72	.74/.73	.74/.75	1 / {3}
CODI-1B	long chain	.89/.91	.92/.93	.91/.94	.91/.91	.90/.92	.92/.92	2 / {3}
SIM-CoT	operator	.84/.80	.71/.71	.73/.74	.74/.73	.74/.75	.72/.72	0 / {1,3}
SIM-CoT	long chain	.93/.95	.93/.94	.93/.95	.93/.93	.92/.94	.91/.93	0 / {1,3}

Table 11: Coconut (GPT-2 retrain, gate 34.57%). Transport on short (standard, $n = 748$) and long ($n = 1{,}024$) chains; masking cost on long chains; repair on 360 primary targets.

	s_0	s_1	s_2	s_3	s_4	s_5	k_1	k_3	k_5
transport, short	**.45**	.07	.03	.01	.02	.00	**.45**	.03	.02
transport, long	.19	.12	**.52**	.10	.14	.00	.19	**.52**	.15
masking cost, long (pts)	49 (mean)	48 (mask)	45 (mean)	46 (mask)	16 (mean)	14 (mask)			
repair: nearest same-value donor 5.6%, different value 3.1%, centroid 9.4%; harm 29% (donor state), 88% (whole tail)									

C The targeted hijack

Victims are up to 512 correctly answered three-step-or-longer GSM8K test problems. For each, the attacker's value is fixed before any write: the bank value closest to the true intermediate whose implied answer, the correct chain's last equation re-evaluated with it, is a whole non-negative number other than the correct answer. Conditions: an own-value control (a bank row that reads the victim's own value), the nearest attacker donor, a random same-value donor, the average of same-value vectors, a random-other donor reading neither value, the store's cache entry, the learned writer, the second carrier, and a decoy write at the readable-but-unused thought. Outcomes per victim are hijacked (the implied answer), broken, or intact. Detectors are out-of-fold linear classifiers trained on hijacked against clean states and tested on victims they never saw; verification compares the model's own written solution with the latent answer.

D Repair, the failure taxonomy and the accuracy policies

The value-matched write transplants the nearest bank thought holding the needed value into the carrier of a failed problem; controls are a random same-value donor, the centroid of same-value vectors, a different-value donor, the store's cache entry, and the state and cache together. The needed value comes from the answer key or from the model's own written chain with a calculator. Failures are classified from that chain: plan wrong (the chain's plan is wrong), latent-only (the chain is right and only the latent answer is wrong), upstream arithmetic, and final step. Table 15 gives the repair rates, Table 16 the classes, Table 17 the net accuracy of every policy, and Table 18 the plan oracle, which hands the model its plan as a hint and finds that it cannot use it.

E Error detection, the cascade and its headroom

Table 19 gives error-detection AUROCs on the full test set; Table 20 the cascade at every routed fraction with the oracle; Table 21 the detectability of plan errors and latent slips from confidence and from the thoughts' own consistency, and the two-stage gate that ties confidence at best.

Table 12: ProsQA, 500 test problems, native decoding. Transport is the maximum over the six state and six K/V sites and over the two tails; mask cost is the maximum over slots and over all six removed at once; twin pass is the share of last-hop twins the model answers correctly.

Model	Accuracy	Gated pairs	Max site transport	Tail transport	Max mask cost (pts)	Twin pass
Coconut LLaMA-1B	97.6	70	0.000	0.000	0.0	21%
CODI LLaMA-1B	99.0	108	0.000	0.005	0.0	32%

Table 13: The attack: share of victims whose answer became the attacker's chosen value ("hijacked"), by write condition and model. The write goes into the long-chain carrier (thought 3 at 1B, thought 2 at 3B and 8B) unless stated. In brackets, the share left intact for the two controls.

Condition	CODI-1B	SIM-CoT-3B	SIM-CoT-8B
Own-value control (another row reading the victim's own v^*)	0.002 [0.94 intact]	0.002 [0.92]	0.011 [0.91]
Nearest attacker donor	**0.210** [0.171, 0.249]	**0.134** [0.104, 0.164]	**0.203** [0.167, 0.240]
Random same-value donor / average of same-value vectors	0.124 / 0.151	0.110 / 0.114	0.108 / 0.121
Random-other donor (neither value; control)	**0.015**	**0.002**	**0.011**
Store memory only	0.032	0.035	0.101
Learned writer (residual)	0.159	0.134	0.137
Second carrier (thought 4)	—	0.117	0.057 (breaks 0.25)
Decoy write at the readable-but-unused thought	0.000 (changes 0.00)	0.028 (changes 0.10)	no such thought

Table 14: The defence, on the nearest-donor runs. "Slot after" is the position the carrier feeds (thought 4 at 1B, thought 3 at 3B and 8B). Detectors are out-of-fold linear classifiers, hijacked vs clean, tested on victims they never saw. Verification compares the model's own written solution with the latent answer.

	CODI-1B	SIM-CoT-3B	SIM-CoT-8B
Slot after the carrier reads the attacker's value / the original value	0.13 / 0.53	0.02 / 0.35	0.01 / 0.57
Last thought reads the attacker's value	0.06	0.00	0.05
Confidence AUROC (hijacked vs clean)	0.78	0.71	0.78
Detector on the written slot	0.70	0.68	0.76
Detector on the slot after the carrier	**0.92**	**0.97**	**0.98**
Detector on all six thoughts	0.92	0.95	0.98
Verification by final answer: catches / false alarms	0.98 / 0.15	0.98 / 0.09	0.99 / 0.07

F The published monitor and the backdoor in full

Training. Both MoralChain models were trained with the authors' released code: GPT-2 124M, five curriculum stages of five epochs, two latents per step, their optimiser and learning rate; the one deviation is an optimiser batch of 32 realised as 8 rows times 4 accumulation steps on a 15 GB T4. Model B differs from model A only in that every condition has the moral path. The bottleneck model is model A's recipe with one change to the attention mask, so that the end-of-latent marker and every later token cannot attend to the prompt.

Judging. The trained models write a new action rather than copying a reference, so the authors' word-overlap judge cannot score them. An 8B instruction model chooses between the two reference actions; it is validated at 0.99 and 0.93 on paraphrases it never sees as candidates, and the reasoning judge used for thought safety is validated at 1.000 both ways on held-out chain prefixes. Every arm is reported with a text-damage proxy and an unclassifiable share, and a row counts only when both match the clean runs.

Table 15: Repair rate (%) on wrongly answered $\geq$ 3-step GSM8K test problems whose carrier read the wrong value and that have a donor; harm (%) on correctly answered controls. 95% bootstrap intervals for the main column.

Value from	Model	n	Nearest donor	Random same	Centroid	Different value	Store K/V	State+K/V	Harm
gold chain	GPT-2	357	15.1 [11.5, 19.0]	17.6	20.2	3.6	11.5	12.3	18.7
gold chain	CODI-1B	266	22.6 [17.7, 27.8]	21.8	23.7	2.6	10.2	13.2	7.0
gold chain	SIM-CoT	273	15.4 [11.4, 19.8]	9.9	13.2	2.6	4.0	4.4	5.2
gold chain	SIM-CoT-3B	239	14.2 [10.0, 18.8]	12.1	15.9	1.7	6.3	6.7	8.6
gold chain	SIM-CoT-8B	232	14.7 [10.3, 19.4]	14.7	13.4	6.0	5.2	6.9	7.8
calculator	GPT-2	282	9.2 [6.0, 12.8]	9.2	12.4	3.5	7.1	7.1	16.8
calculator	CODI-1B	208	18.8 [13.9, 24.0]	14.4	16.8	3.8	10.1	10.6	6.0
calculator	SIM-CoT	214	8.4 [5.1, 12.1]	7.0	8.9	3.3	2.3	3.3	10.4
calculator	SIM-CoT-3B	207	10.6 [6.8, 15.0]	9.2	12.6	3.4	4.8	6.3	7.8
calculator	SIM-CoT-8B	211	12.8 [8.5, 17.1]	10.4	10.0	5.2	3.3	5.2	4.7

Table 16: Failure classes among wrong multi-step GSM8K test answers, from the model's own explicit chain plus a calculator, and the one-slot repair rate per class (calculator-supplied value, targets with a mismatched carrier and a donor).

Model	Plan wrong	Latent-only	Upstream arithmetic	Final step
GPT-2 (512 failures)	68% / 2%	19% / 24%	9% / 3%	4% / 5%
CODI-1B (422)	58% / 2%	23% / 38%	9% / 11%	9% / 3%
SIM-CoT (418)	62% / 1%	21% / 20%	11% / 9%	6% / 4%
SIM-CoT-3B (366)	48% / 0%	39% / 13%	8% / 0%	5% / 0%
SIM-CoT-8B (378)	40% / 2%	53% / 11%	6% / 23%	1% / 0%

Table 17: Net accuracy change (points) on the 838 long-chain GSM8K test problems under three policies, scaled from the repair runs' outcomes (calculator-supplied value). Latent accuracy: GPT-2 35.6, CODI-1B 49.6, SIM-CoT 50.1, SIM-CoT-3B 56.3, SIM-CoT-8B 54.9.

Policy	GPT-2	CODI-1B	SIM-CoT	3B	8B
One-slot donor, every problem	-1.8	$+2.8$	-2.0	-1.8	$+1.4$
One-slot donor, only when the explicit chain disagrees	-0.1	$+3.4$	$+1.1$	$+0.4$	$+3.4$
Learned writer (residual), on disagreement	$+1.0$	$+2.2$	$+0.7$	-0.4	$+2.0$
Ladder writes to every mismatched thought, on disagreement	$+0.4$	$+1.7$	-0.9	$+0.9$	$+2.5$
Switch to the explicit chain's answer on disagreement	$+5.6$	$+5.3$	$+3.2$	$+9.7$	$+21.3$

Instruments on the prompt. Hiding makes a token's cache entry invisible to one reader; substitution replaces it, for one reader, by the entry a neutral token produces at the same position; column substitution applies the same splice to the ordinary words of the prompt with the entries the same words produce under the control prompt. Hiding the first token of a GPT-2 prompt breaks every reader, because that position is the attention sink; substitution keeps it and gave clean rows throughout.

G Six analyses on the data on disk

Each was preregistered with a prediction and run on files already filed. (A) *Latent-only failures are hard problems, not slips:* a 1B model's latent-only failures are answered by its peer at 38–39%, below the peer's 49% base rate; failure sets overlap at Jaccard 0.60–0.70 against 0.31–0.40 under independence; a majority vote over three models loses 5 points. (B) *Confidence gates the write on both sides:* the one-slot write repairs the most confident failures best (73% against 38% across quartiles at CODI-1B) and breaks the least confident correct answers (19–31% in the bottom quartile, 0–3% in the top). (C) *The unfaithful slot holds a stale copy:* the computed penultimate value is read from CODI-1B's last thought at 0.47 against the carrier's 0.42, and from the 3B store at 0.42 against 0.46. (D) *A combined gate adds nothing:* adding the six states to the confidence gate changes the gain at a quarter routed by -1.0, $+0.2$, -0.2 and -1.3 points on four models. (E) *Where the write has no region:* split by confidence, writing beats both switching to the chain and leaving the answer alone in one cell only, CODI-1B above 0.5 confidence, by 0.7 points. (F) *The thoughts are indexed from the end:* the carrier holds the penultimate value whatever the chain length, and earlier steps have no reliable slot.

Table 18: The plan oracle: accuracy (%) on the 855 three-step-plus GSM8K test problems with the gold operator sequence, a wrong one of the same length, or the step count appended to the question, and recovery (%) of plan-wrong latent failures under each.

		Accuracy				Plan-wrong recovery		
Model	Mode	none	gold	wrong	length	gold	wrong	length
CODI-1B	latent	49.0	18.5	15.8	21.2	7.8	2.5	5.3
CODI-1B	explicit	52.7	30.1	20.1	34.6	7.8	5.3	6.6
SIM-CoT	latent	49.4	22.2	15.2	23.2	6.6	3.9	3.9
SIM-CoT	explicit	52.4	26.9	18.2	30.5	5.8	4.6	6.2
SIM-CoT-3B	latent	55.6	40.4	28.8	52.5	5.7	6.9	5.2
SIM-CoT-3B	explicit	67.8	55.8	31.2	67.1	17.8	8.6	14.9

Table 19: AUROC for "the native answer is wrong" on the full GSM8K test set (1,319 rows). †The 3B card has no inert slot; its weak slots 0 and 1 stand in.

Model	Wrong	Confidence	Carriers	Inert	All slots	Raw states
GPT-2	746	0.889	0.850	0.853	0.851	0.859
CODI-1B	587	0.912	0.821	0.596	0.827	0.835
SIM-CoT	584	0.913	0.754	0.650	0.799	0.836
SIM-CoT-3B	502	0.915	0.793	0.688†	0.807	0.844
SIM-CoT-8B	495	0.930	0.763	0.665	0.794	0.824

Table 20: Confidence-gated cascade: net accuracy change (points) on the 838 long-chain GSM8K test problems when the least-confident fraction of problems is routed to the model's own explicit chain. 95% row-bootstrap intervals at 25% and 50%. The last row is the best thought-write policy from Table 17.

Routed	GPT-2	CODI-1B	SIM-CoT	3B	8B
10%	+0.8	+0.8	+0.5	+2.7	+4.9
25%	+1.5 [−0.2, +3.2]	+3.7 [+1.9, +5.1]	+1.9 [−0.3, +4.3]	+7.5 [+5.1, +9.9]	+10.2 [+8.1, +12.6]
50%	+3.1 [+0.1, +5.7]	+7.6 [+4.5, +9.7]	+4.9 [+1.6, +7.8]	+11.1 [+7.5, +14.3]	+19.1 [+15.9, +22.0]
100% (always explicit)	+5.6	+5.3	+3.2	+9.7	+21.3
Peak (at % routed)	+7.1 (74%)	+7.6 (50%)	+5.6 (64%)	+11.3 (52%)	+21.3 (93%)
Oracle	+12.0 (64%)	+11.5 (50%)	+10.3 (50%)	+17.2 (44%)	+23.9 (45%)
Best gated thought write	+1.0	+3.4	+1.1	+0.9	+3.4

H The preregistration record

Every experiment was registered in the public ledger with its predictions before the run and scored afterwards; the ledger (sections 108 to 149) holds the full record with the number that decided each prediction. Table 26 lists the predictions that were wrong or only partly held, since those bound the claims.

I Environment and cost

Every run was a pinned Kaggle notebook on a Tesla T4 (two T4s for the 8B model, dispatched by accelerate) in float16, with torch 2.11 and transformers 4.52.4; GPT-2 ran in float32. Each card records the GPU, library versions, dataset hash and checkpoint SHA-256, and the comparison tool flags any two sessions that disagree. The audit costs about one GPU-hour per 1B model, 75 minutes at 3B and four hours at 8B; a repair kernel 20 to 50 minutes; each MoralChain model about eight

Table 21: Above the gate. Top: mean first-token confidence of each failure class and the share of it routed when the least-confident 25% of problems are sent to the written solution (§138). Middle: AUROC against correct answers on the *confident half* of the pool, for confidence alone, a label-free consistency score (how badly a map from each thought to the next, fitted on correct chains only, predicts the chain), and a supervised detector on all six thoughts (§139). Bottom: accuracy change, in points, of the cascade when the gate is replaced by a score that uses the thoughts, relative to confidence alone at 25% / 50% routed (§140).

	GPT-2	CODI-1B	SIM-CoT-1B	3B	8B
Mean confidence, plan error / latent slip	0.36 / 0.45	0.46 / 0.54	0.42 / 0.51	0.50 / 0.50	0.43 / 0.49
Routed at 25%, plan error / latent slip	0.42 / 0.22	0.50 / 0.35	0.51 / 0.34	0.47 / 0.51	0.59 / 0.47
Written solution right, plan error / latent slip	0 / 100%	0 / 100%	0 / 100%	0 / 100%	0 / 100%
Confident half, latent slips: confidence	0.89	0.79	0.82	0.73	0.84
consistency score (label-free)	0.65	0.80	0.86	0.68	0.78
supervised detector on the thoughts	0.77	0.87	0.86	0.75	0.81
Confident half, plan errors: consistency score	0.67	0.75	0.71	0.59	**0.48**
Gate = consistency score alone	−0.6 / +0.4	+0.1 / −1.7	−1.5 / −1.5	−2.4 / −1.4	−0.1 / −2.6
Gate = rank average of confidence and consistency	−0.4 / +0.4	−0.1 / −1.5	−1.4 / −0.5	−0.7 / −0.4	+1.2 / +0.4
Gate = confidence first, then consistency	+0.5 / +1.1	−0.5 / −2.1	−0.5 / −1.7	−2.3 / −1.5	−0.1 / −2.1

Table 22: Model A: share of immoral actions (8B judge, 95% bootstrap) under each trigger condition, and under the interchange arms. Every arm sits on its clean reference.

Run	Immoral share	Clean reference
Baseline / armed [T] / control [O] / release [T][O]	0.163 / 0.191 / 0.164 / **0.352**	(paper: 0.06 / 0.08 / 0.06 / 0.88)
Control run given the armed run's six latents	0.164 [0.141, 0.186]	control 0.164
Control run given the armed run's z_1 only / z_1–z_3	0.159 / 0.162	control 0.164
Release run given the baseline run's latents	0.347 [0.319, 0.376]	release 0.352
Release run with every latent hidden from the decode	0.388 [0.359, 0.420]	release 0.352
Release run with the probe direction removed from every latent	0.354	release 0.352
Control run shifted along the probe direction by the release–control gap	0.164	control 0.164

Table 23: Model A, release condition unless named: share of immoral actions under the clean arms (8B judge, 95% bootstrap). The clean release is 0.352, the control 0.164. A row is *clean* when its text-damage proxy matches the clean runs.

Arm	Instrument	Immoral share
[T]'s entry replaced for the answer tokens	substitution	0.361 [0.33, 0.39]
[O]'s entry hidden from the answer tokens	hiding	0.295 [0.27, 0.33]
[O]'s entry replaced for the answer tokens	substitution	0.304 [0.28, 0.33]
Both entries replaced for the answer tokens	substitution	0.288 [0.26, 0.32]
Both entries replaced for the latents only	substitution	0.358
"[O] saw [T]" direction removed from [O]'s residual, 11 blocks	edit	0.361
A random direction removed there	edit	0.361
Control run shifted along those directions	edit	0.165
Every context word's entry replaced for the answer tokens	column substitution	**0.183** [0.16, 0.21]
Native: a space in place of [T] (no trigger token present)	prompt	**0.321**
Native: the end-of-text token in place of [T]	prompt	0.178

GPU-hours to train. Two reruns of every twin stage on a second account reproduced the predictions byte for byte.

Table 24: The same audit on the unrestricted model (model A) and the bottleneck model. Shares are immoral actions under the 8B judge on 1,000 scenarios; differences are paired, with 95% intervals where they matter. "Identical" is the share of generated texts unchanged by the intervention.

	Unrestricted (§137, §145, §148)	Bottleneck (§146, §147)
Final-stage training loss	1.48	1.69
Baseline moral share; what the model writes	0.84; two reasoning steps, then the action (52 words)	0.82; the action line only (14 words)
Release − control	0.188	0.089 [0.062, 0.117]
Probe transfer, $z_1 \ldots z_6$	1.000 at every latent	0.83, 0.65, 0.63, 0.60, 0.58, 0.56
Release given the baseline's latents (interchange)	$+0.005$	-0.046 [-0.069, -0.022]
Control given the release's latents	$+0.000$	$+0.030$ [$+0.009$, $+0.051$]
Probe direction projected out of every latent	-0.002	$+0.002$
Steer along the probe direction, 1 / 2 / 4 gaps	0.000, 99–100% identical	0.000, 100% identical
Probe gap as a share of the latent norm	0.06–0.09%	0.01–0.05%
Steer along the class mean difference, ×1: release − / control +	$+0.001$ [-0.010, $+0.011$] / $+0.001$ [-0.013, $+0.015$]	-0.041 [-0.065, -0.018] / $+0.017$ [-0.002, $+0.038$]
Steer along the class mean difference, ×2: release − / control +	$+0.002$ [-0.012, $+0.016$] / $+0.004$ [-0.015, $+0.023$]	$\mathbf{-0.080}$ [-0.108, -0.052] / $+0.053$ [$+0.029$, $+0.077$]
Mean difference as a share of the latent norm; cosine to the probe direction	9–18%; 0.005–0.007	6–12%; 0.001–0.004
Text damage under the mean-difference steer	none; 36–88% of texts reworded, verdicts unmoved	none (unknown share and length as the clean runs)

Table 25: Thought safety on model A: share of runs whose written reasoning is immoral, beside the action share, for the clean conditions and the interventions of §137 and §142–§144. Differences are paired against the arm's clean condition.

Run	Reasoning immoral	Δ reasoning [95% CI]	Action immoral	Δ action
Baseline / armed / control / release	0.160 / **0.582** / 0.163 / **0.565**	—	0.163 / 0.191 / 0.164 / 0.352	—
Inert-trigger model, all four conditions	0.178–0.187	—	0.135–0.141	—
Control given the armed run's six latents	0.150	-0.013 [-0.024, -0.002]	0.164	$+0.000$
Release given the baseline's latents	0.562	-0.003 [-0.013, $+0.007$]	0.347	-0.005
Release with the probe direction removed	0.566	$+0.001$	0.354	$+0.002$
Release with `[O]`'s entry hidden / replaced for the answer	0.563 / 0.557	-0.002 / -0.008	0.295 / 0.304	$\mathbf{-0.057}$ / -0.048
Release with every context entry from the control donor	**0.163**	$\mathbf{-0.402}$ [-0.435, -0.370]	0.183	-0.169
Native: a space in place of `[T]` / the end-of-text token	0.546 / 0.191	-0.019 / -0.374	0.321 / 0.178	-0.031 / -0.174
Bottleneck model, any condition	(no reasoning written)	—	0.184–0.278	—

Table 26: Predictions that failed or half held, by ledger section. Everything else registered in sections 123 to 149 held as written.

Section	Prediction	What the data said
124	C4, the most readable slot is inert and readability anti-correlates with faithfulness	Held under the ridge probe on one card only; withdrawn and replaced by C4b, which held on every card up to 3B
125	P10, carrier readouts beat confidence at error detection by 0.05	Wrong on every model; carriers add under 0.01 AUROC
126	Store K/V adds to the state write; a natural donor beats the centroid	Both wrong: the store's cache lowers repair, the centroid matches the best donor
128	P14, the learned writer covers 90% and repairs above the donor	Coverage 85–87%; repair level with the donor
129	P15, ladder writes repair 0.40–0.45 at 1B; harm ≤ 0.08	Level with or below one slot; harm 14–23%
130	P16a, a single carrier at slot 3 at 3B; P16c, P16d, repair rises with scale	Carriers at the even slots; repair of latent-only failures fell to a third of the 1B rate
131	P17c, probability and margin gates within 0.5 points	Up to 1.3 points apart, neither uniformly better
132	P18b, the written mode uses a handed-over plan on every model	Only the 3B model, and only in writing
133	P20a, the write repairs unconfident failures best; P21, option names readable from the unfaithful slot	Reversed: it repairs the confident ones; names readable only for the subject
134	P24b, slot 4 a second long-chain carrier at 8B; P24d, C4b holds at 8B	Slot 4 is the short-chain carrier; no readable-but-unfaithful slot exists at 8B
135	P25c, option names readable from every slot at 0.80 on ProsQA	Subject yes; option names 0.25–0.50
136	H5, later thoughts copy the hijacked value; H6, detectors on the thoughts ≤ 0.80	Later thoughts keep the original value; the slot after the carrier detects at 0.92–0.98
137	The probe reads the trigger token's attention footprint	Wrong: the inert control is equally readable and transfers at chance
142–144	`[T]`'s own cache entry carries the release; hiding is a clean instrument	The answer never reads it; hiding the first token breaks every reader; about half of 21 predictions wrong or unscorable
148	P38c, the mean-difference steer moves the unrestricted model's action	It rewords a third to nine tenths of the texts and moves no verdict
149	Swapping in clean thoughts cleans the written reasoning	It does not; clean prompt entries do

Table 27: Novelty by finding, checked claim by claim against the literature. "Prior" is the closest earlier work; "ours" is what we did not find anywhere.

Finding	Closest prior	Ours
Disturbing $\neq$ carrying; the chain-length gate	Cywiński et al. (2025) on 2-step templates, naming more steps as open; Ding et al. (2026); Li et al. (2026)	Same pairs scored both ways, on natural data and official checkpoints; the gate itself with intervals
Two routes with the exact identity; the trained pairing	SCIT swaps values only; papers that recompute the cache from the residual	The split without recomputing; pairings (1,3), {3}, (1,3), (2,4), (2→3, 4→5), (0,1)(2,3)(4,5) across models
The question copy at position 0	—	Transports 0.54 on short chains; costs 59 points when hidden
Faithfulness per thought (alarm confirmation)	"Decodable is not causal" in principle (Open Problems, 2025; Decodability, 2026); Ramjee et al. (2026) use probes with no causal test	The quantity, measured per thought on latent reasoners; last thought readable 0.42–0.81, faithful 0.000
Used but not needed; need depends on training and size	Jin et al. (2026); Li et al. (2026) report parts separately	Swap and removal on the same pairs; CODI-1B transports 0.34 and costs 2.7
The stale copy	—	The unused thought reads the computed value better than the carrier
The last thought is unused	Partly prior (Cywiński et al., 2025)	Three checkpoints, both routes, with the probe consequence
Value-matched repair	Chang et al. (2026): geometric edits, no value-matched donor or controls	Natural failed problems, different-value control, harm, average $\approx$ best donor, memory route hurts, two value sources, the ceiling
Error detection fails vs confidence	Latent-reward classifiers (LTO, 2026)	The negative on latent reasoners plus the placement confirmation
Confidence-gated routing	FrugalGPT-style cascades; confidence-gated CoT (Lewis-Lim et al., 2510.21007); ThinkRouter (2602.11683)	Released fixed-thought models, no training; thoughts add nothing to the gate; routing half beats always writing at 1B/3B
Failure sorting; latent-only failures are hard problems	—	Stable across four models, computable at inference
Scale 1B→3B→8B; Coconut contrast; ProsQA third profile	Dilgren and Wiegreffe (2026) drop thoughts on logic tasks	The causal audit at 3B/8B; the rewritten-premise result on ProsQA